

Utkarsh Singh Gunjiyal

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Education

Delhi Technological University, BTech in Computer Science
Ishwar Singh Dev Samaj Sr. Sec. School, Chandigarh, PCM

Sep 2022 – May 2026
March 2022

Experience

Business and Data Analytics Intern, TATA Motors Jamshedpur Plant

Sep 2025 – Nov 2025

- Built and deployed a Power BI dashboard with 12+ real-time visualizations monitoring key supply chain KPIs, reducing manual tracking time by 40%.
- Designed ETL pipelines over 100,000+ rows of raw logistics data with structured validation and transformation workflows to improve downstream analytics accuracy.

AI Engineer Intern, The Future University – Chandigarh, IN

May 2025 – Jul 2025

- Developed an institutional-grade AI stock scoring system evaluating 5000+ Indian equities using multi-pillar fundamental analysis and narrative risk sentiment; automated end-to-end via n8n agentic workflows and Gemini LLMs.
- Built 5+ agentic trading workflows with LLM-based multi-step reasoning for chart analysis, risk assessment, and trade evaluation; integrated Promptfoo for LLM evaluation and PostHog for observability, reducing manual steps by 30%.
- Deployed ChatGPT-powered Chart Analyser and Portfolio Analyser with structured prompt logic and iterative evaluation pipelines, serving 4,000+ active users.

Projects

Smart Traffic Monitoring Multi-Object Tracking System

- Developed a computer vision pipeline using YOLOv8, OpenCV, and SORT tracking for real-time vehicle detection and monitoring.
- Implemented Kalman Filter-based object tracking to maintain stable vehicle identities across video streams.
- Built event detection and vehicle counting logic using trajectory-based line crossing algorithms.
- Optimized inference and processing pipelines for near real-time performance.

Full-Stack AI Assistant Platform (React, FastAPI, MongoDB)

GitHub | Live Demo

- Designed and developed a full-stack AI assistant platform using React, FastAPI, MongoDB, and JWT authentication supporting persistent user conversations.
- Implemented thread-based conversation management, message persistence, and role-based chat history storage using MongoDB collections.
- Built hybrid retrieval pipelines using FAISS and BM25 search with semantic caching to improve response quality and reduce repeated inference costs.
- Developed modular REST APIs for chat, authentication, document upload, retrieval, and conversation history management.
- Implemented JWT-based authentication, thread-aware memory, document retrieval, and multi-user session management workflows.

Document Processing & Approval Workflow System

- Built a backend workflow automation platform using FastAPI and MongoDB for document ingestion, validation, and approval management.
- Developed PDF extraction pipelines using PyMuPDF and rule-based validation engines for automated document verification.
- Implemented human-review workflows for low-confidence cases with status tracking and approval audit trails.
- Designed scalable REST APIs and modular service architecture enabling automated processing of large document volumes.

Skills

- Languages:** Python, SQL, JavaScript, C++
- Frontend:** React, HTML, CSS
- Backend:** FastAPI, REST APIs, JWT Authentication, Async Programming
- Databases:** MongoDB, SQL, FAISS Vector Database
- ML & AI:** PyTorch, Scikit-Learn, CNNs, NLP, Retrieval-Augmented Generation (RAG), Vector Embeddings
- Tools & Technologies:** Git, GitHub, LangGraph, LangSmith, Promptfoo, PostHog, Power BI, n8n
- Data Processing:** Pandas, NumPy, ETL Pipelines, Data Validation & Transformation